

FLAMINGO Hybrid V2 Cave Radio

Assembly Instructions

Tools

- Hobby knife
- Wire cutters/diagonal cutters
- Wire strippers (down to 28 AWG)
- Soldering iron
- RTV or hot glue
- 1/4in heat shrink tubing
- 8mm deep well socket (bit only)
- Pliers
- Small Phillips head screwdriver
- 2mm hex driver
- 2-part epoxy [Optional]
- Digital multimeter [Optional]



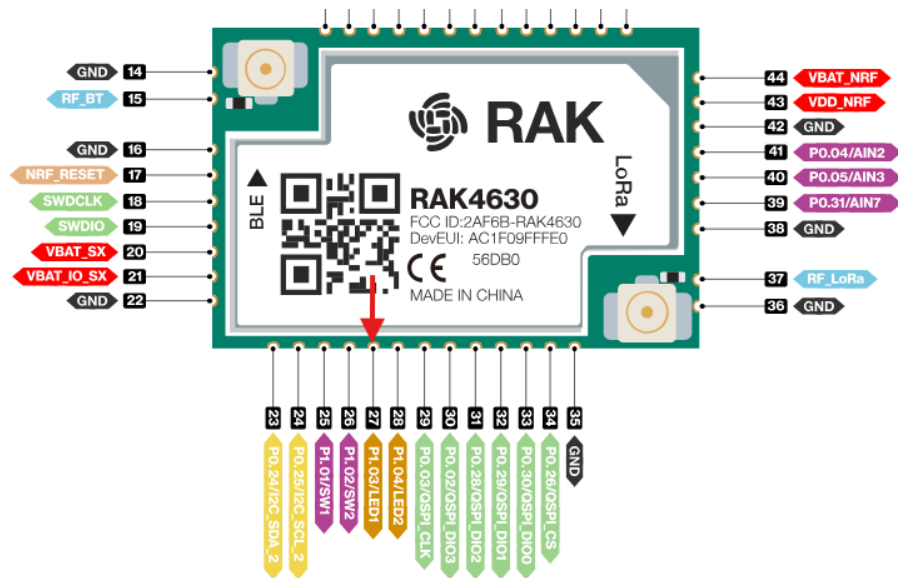
FDM Parts

1. Print 1 each of the **Main Housing** and **Top Lid** parts using high-strength material like PETG or ABS
 - a. Recommend 50% minimum infill
 - b. If possible, use ironing on top surfaces (i.e. to provide smooth surfaces for gaskets)
2. Print 1 each of the **Battery terminal** sides **A** and **B** out of any rigid plastic material
3. Print 1 each of the **Gasket** and **Mud Flat parts** out of soft TPU
 - a. Use ironing if possible
 - b. Use high quality (thin layers)
4. Deburr all parts (especially holes and interfaces) with hobby knife as needed

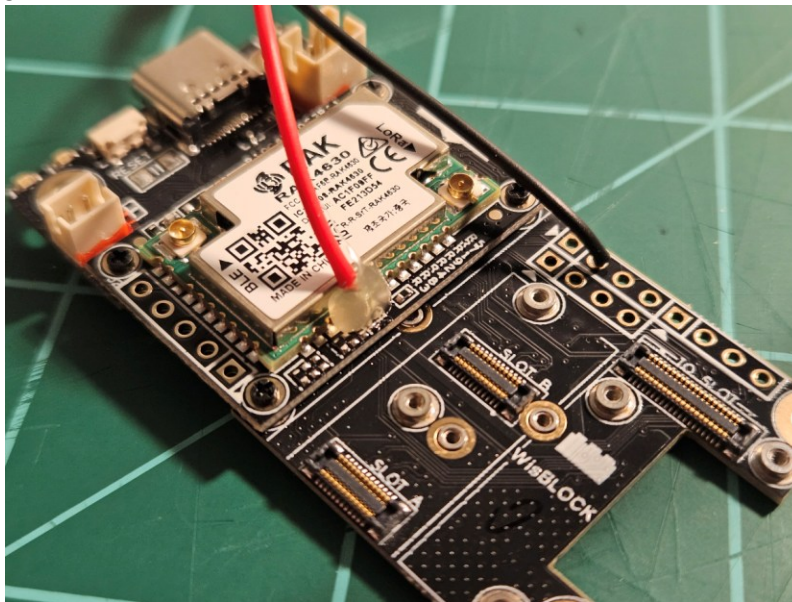
Electronics Preparation

1. Cut prewired green LED to ~3.5in; strip and tin last 1/8in
2. Carefully solder positive lead onto RAK4630 LED1 pin*; pot with RTV or hot glue

***NOTE:** Due to conflicts with using the RAK5802, the preferred GPIO 1 or 2 pins are *not currently available* for running the external LED on the hybrid nodes. Follow the Github for updates, otherwise either skip this step or be careful soldering.



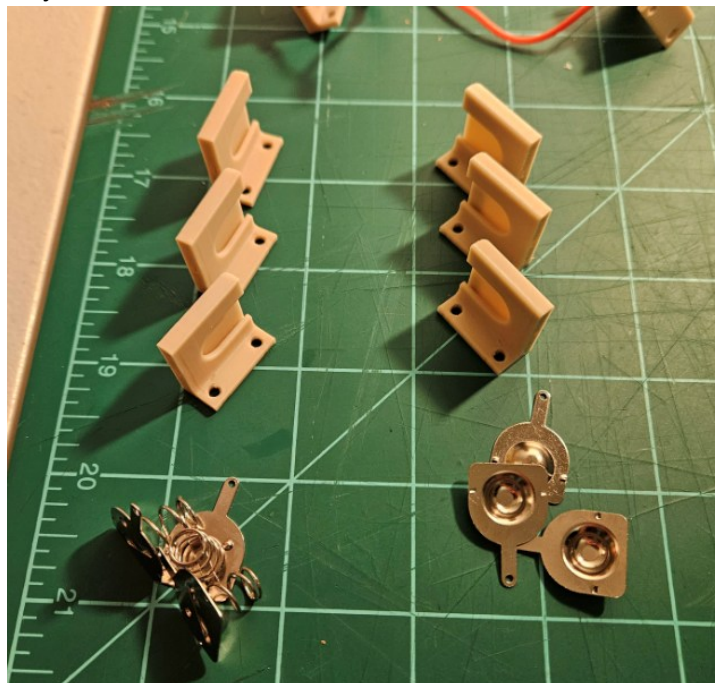
3. Solder negative lead onto a GND pin on GPIO



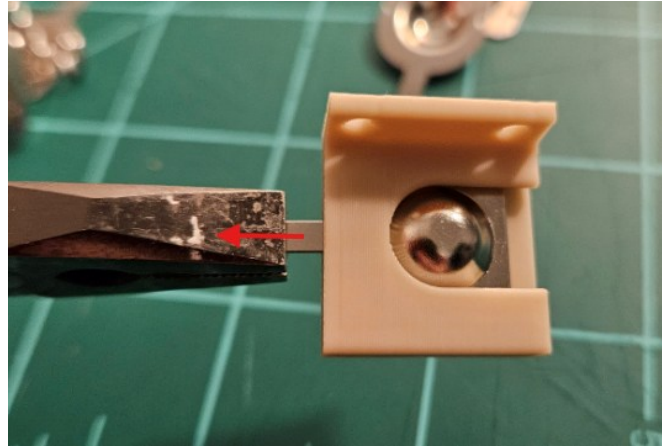
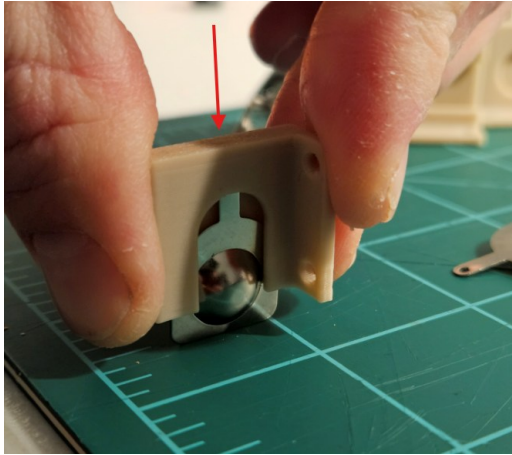
4. Cut leads of waterproof pushbutton down to ~1in; strip and tin last 1/8 in
5. **Verify polarity of JST-PHR-02 pigtail!** These often ship reversed
6. Cut leads of JST-PHR-02 pigtail down to ~1in and keep red scrap; strip and tin last 1/8 in
7. Strip and tin last 1/8 in of red tail and solder onto one lead on pushbutton as shown; apply 1/4in heat shrink tubing
8. Apply heat shrink tubing and solder positive lead of JST-PHR-02 onto other pushbutton lead as shown



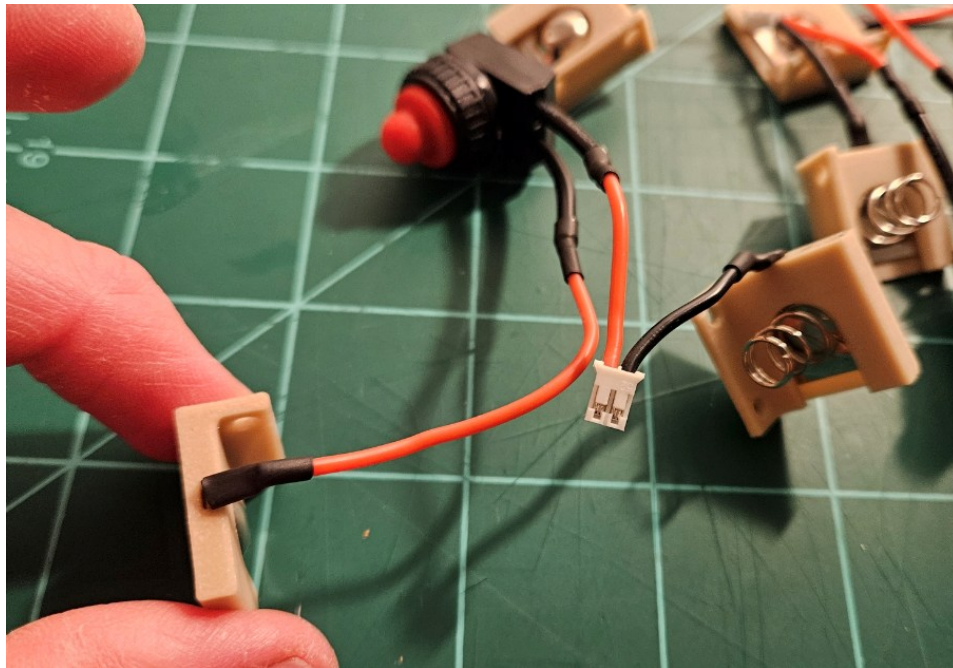
9. Match up battery terminals with their FDM holders as shown



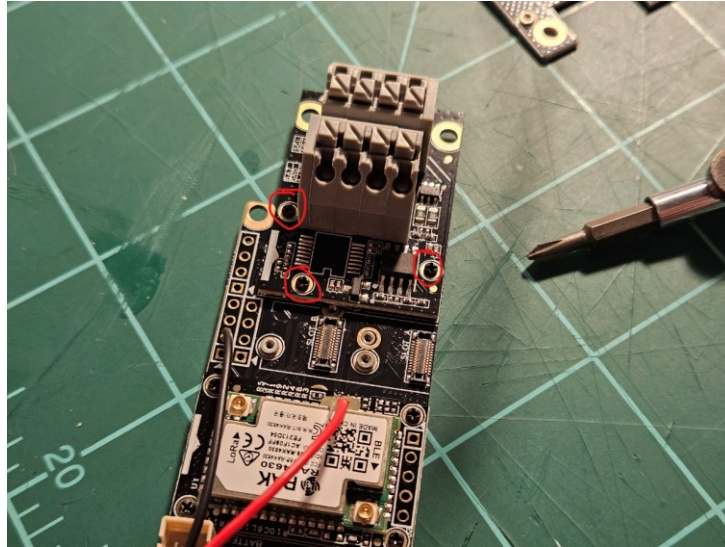
10. Press-fit each terminal into their holders
- Start by lining up tab and pressing against flat surface
 - Use pliers to pull tab until terminal is fully seated in holder (will bottom out)



11. Apply 1/4in heat shrink and solder the red harness lead onto the positive (smooth) terminal's tab
12. Apply 1/4 in heat shrink and solder the black harness lead onto the negative (spring) terminal's tab
13. Fold the tabs over toward the inside as shown



14. Attach RAK5802 and secure with 3 small (M1.6?) screws



15. Cut one red and one black alligator test lead in half (or to desired length); strip and tin the last $\frac{1}{4}$ in of each

Radio Assembly

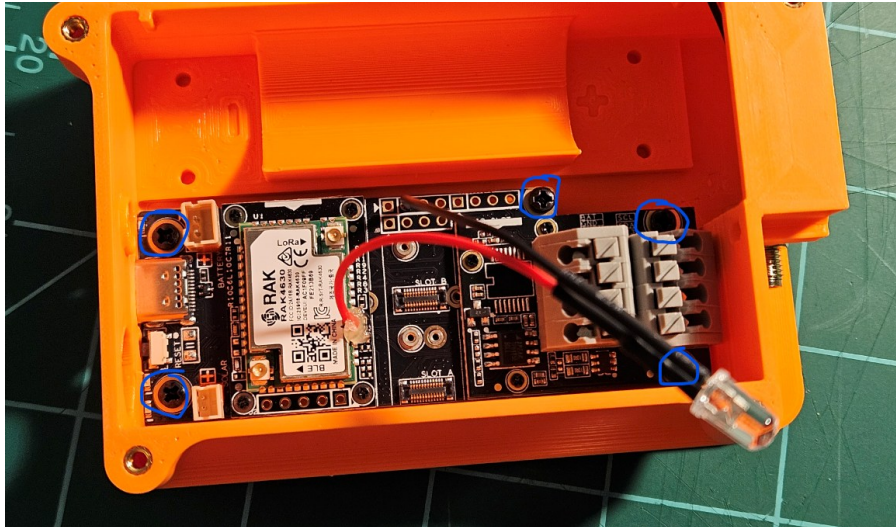
1. Insert 4x M2.5 brass heat set threaded inserts (narrow side down) into each of the four mounting hole locations using a soldering iron until flush
 - a. Recommend setting it to $\sim 400^{\circ}\text{F}$ if using PETG



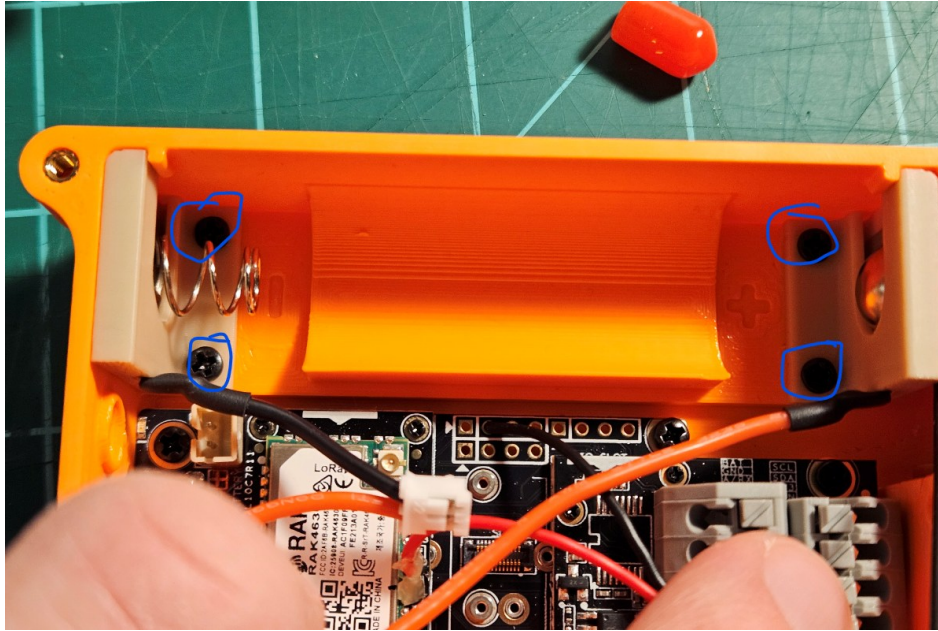
2. Feed SMA-U.FL adapter outward all the way through SMA hole in enclosure
3. Attach crush washer and nut and hand tighten until snug. A deep well 8mm socket should work for this



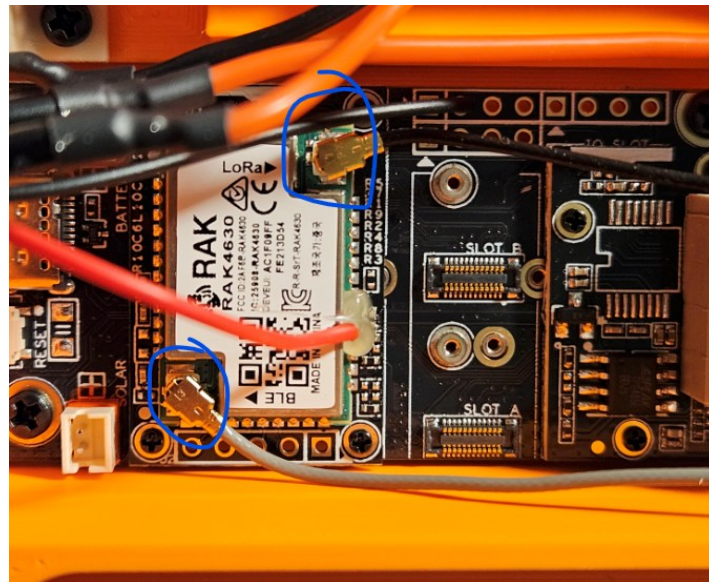
4. Insert radio assembly into main housing and secure with 5x short M2.5 screws



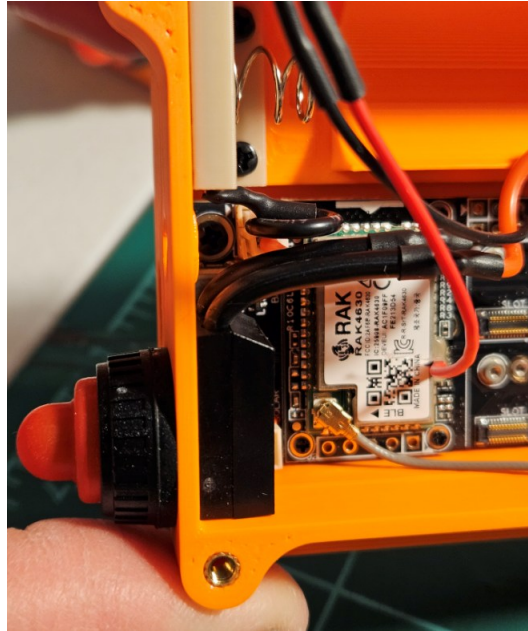
5. Attach battery terminals using 2x short M2.5 screws each. NOTE the polarity indicators embossed on the enclosure.



6. QUALITY CHECK: Temporarily attach 18650 battery and toggle pushbutton to verify functionality
 - a. Set pushbutton to **OFF**
 - b. Remove 18650 battery
7. Plug in the 900MHz and BLE antennas into their respective U.FL ports on the RAK4630



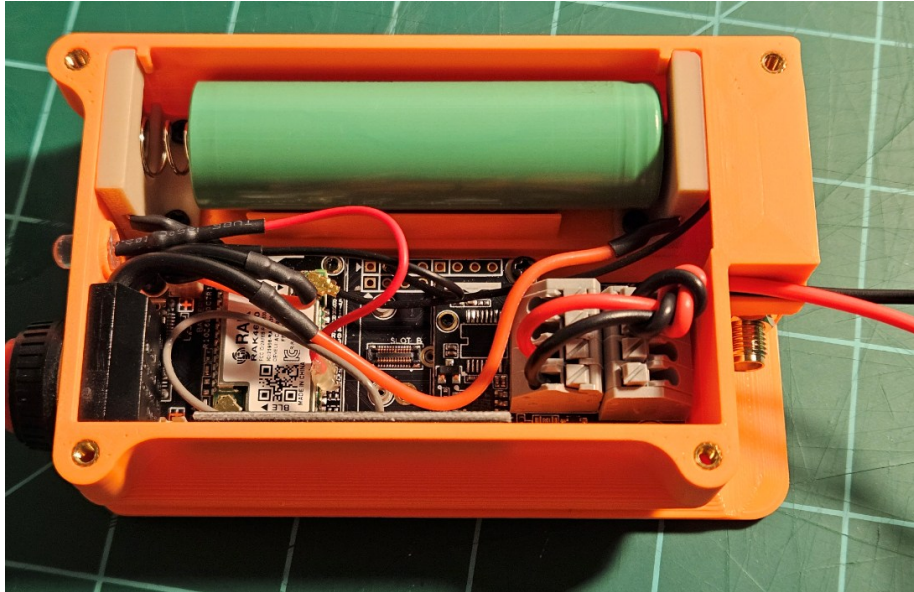
8. Fold the leads of the pushbutton down and carefully plug the JST-PHR-02 plug into the RAK19007 BATT socket
9. Attach pushbutton into front wall and capture by threading on O-ring and pushbutton cap
 - a. Sandwich the mud flap underneath if needed



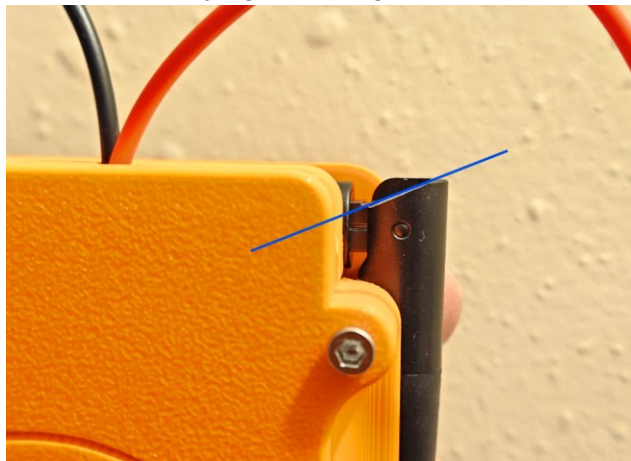
10. Attach BLE antenna onto inner wall using included adhesive as shown
11. Press fit LED bulb into front wall as shown
 - a. Use 2-part epoxy for waterproof seal
12. Thread alligator pigtails in through the back of the enclosure as shown
13. Tie and tighten a dressed double overhand knot ~1 ¼ in from the exposed leads



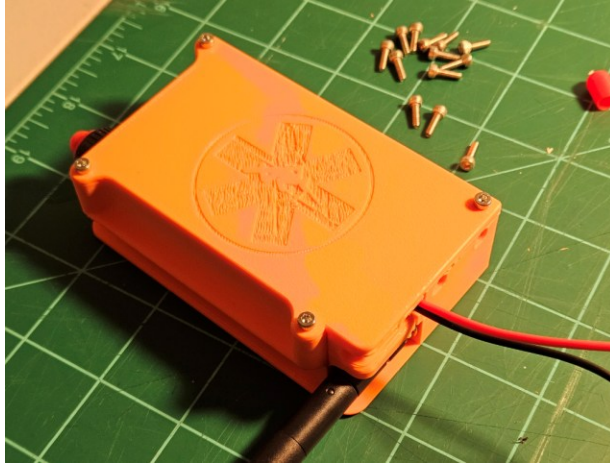
14. Plug the alligator leads into the RAK1802 terminal block in the positions shown
 - a. Pot alligator leads with 2-part epoxy for waterproofing
15. Insert 18650 battery



16. Use diagonal wire cutters to trim rear edges on both sides of folding antenna housing as shown to allow better clearance when folding
17. Thread on antenna and carefully tighten snug with pliers



18. Attach lid using 4x M2.5 x 8mm screws and hand tighten until snug using 2mm hex wrench



Assembly complete.

Refer to Flamingo Operating Instructions on how to install software.